

OPTIMIZING TRAJECTORY COMPUTATION IN MICRO DRONES

ILLINOIS TECH

Armour College of Engineering

Emiliano Mora, Nelson Rosa

Department of Mechanical, Materials, and Aerospace Engineering

Motivation and Background

Task automation remains a key focus in industries such as healthcare, manufacturing, and especially in military applications. A unique challenge in these spaces is the limited resources to compute optimal decision policies.

The long-term goal of this project is to determine the minimum hardware footprint to perform autonomous model-predictive control algorithms.

Model Predictive Control (MPC) offers a framework for real-time trajectory correction

- TinyMPC: Efficient but limited to simplified systems
- ACADOS: Powerful but high storage cost and complex interfacing

Bridging the gap: Provide robust control of complex systems that fit within hardware constraints

Objective

Program a Crazy Flie 2.1+ micro drone to execute a figure 8 trajectory while logging state estimates and reference trajectory

Hardware Buildout

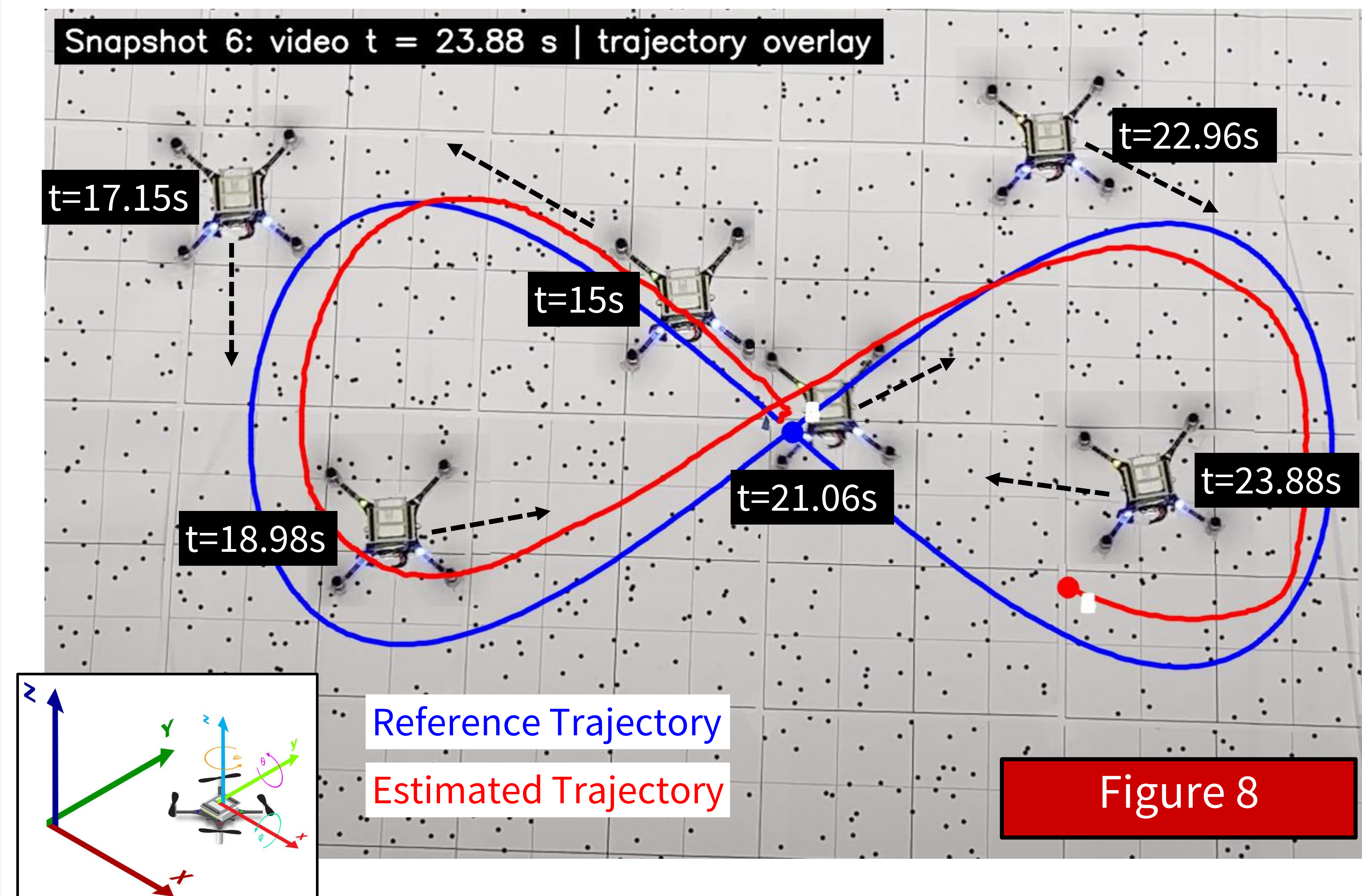
Crazy Flie 2.1+

- STM32F405: Main controller
- nRF51822: Radio and power management

Flowdeck V2 (Upgraded sensor deck)

- Time of Flight sensor (VL53L1x)
- Optical Flow Sensor (PMW3901)

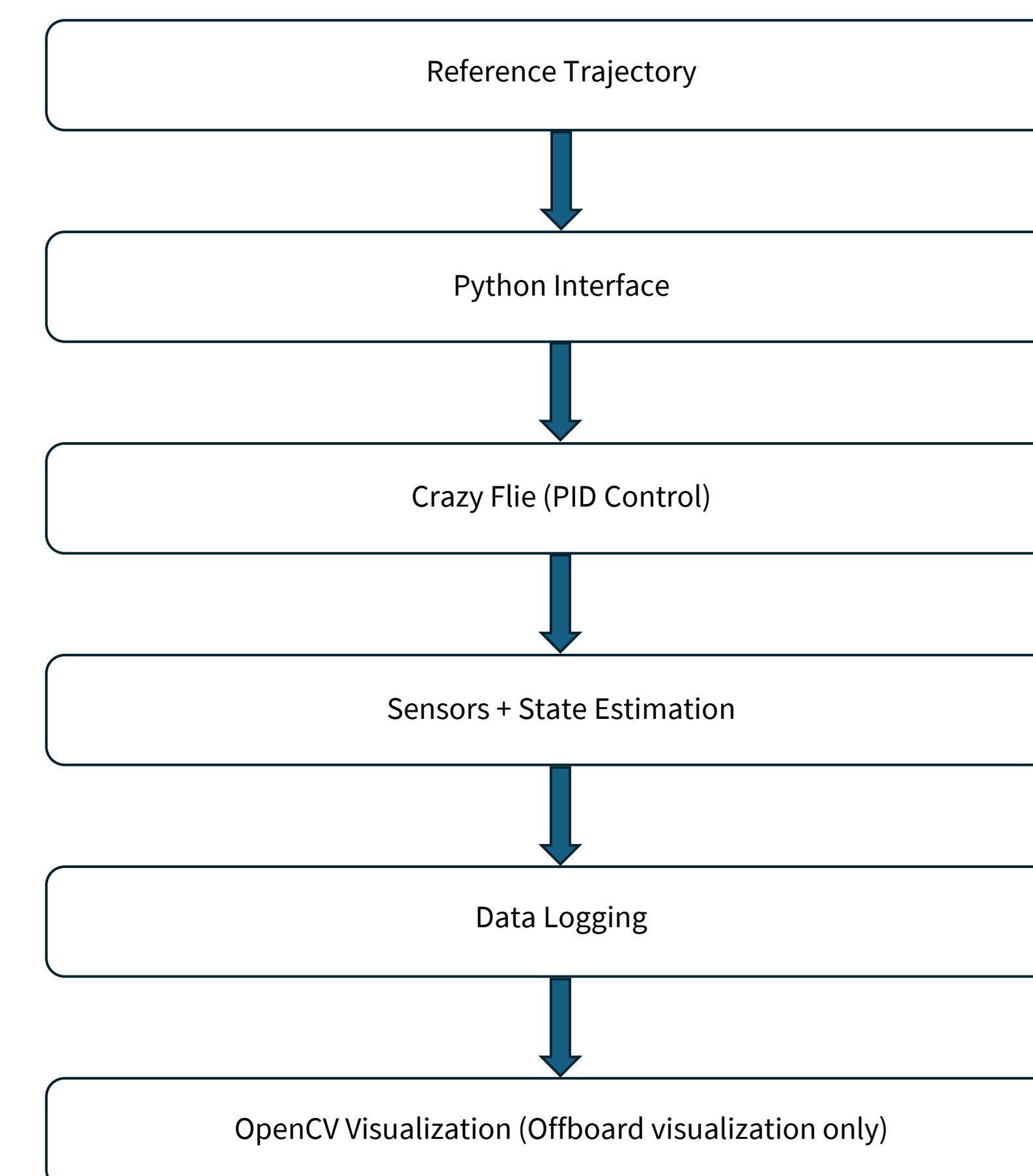
Methods and Approach



Pictured Above: Reference trajectory (blue) vs estimated position (red), reconstructed from logged flight data and visualized using OpenCV
***Note: Timestamps in reference to video recording, not data logs**

Summary

This work demonstrates successful figure 8 tracking on a micro drone using onboard state estimation. Python-based interfacing enabled command execution and data logging, while OpenCV was used to generate scaled visualizations of reconstructed trajectories. This establishes a foundation for future ground-truth validation, testing of more aggressive maneuvers, and exploration of alternative MPC frameworks.



Future Work

Tracking Accuracy:

- Refine OpenCV pipeline and reduce estimation noise

Validation:

- Integrate motion capture for ground truth comparison

Control:

- Evaluate alternative MPC frameworks for embedded systems

Implementation:

- Transition toward fully onboard processing

References

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